PatchFerret user guide

PatchFerret turns a mixing console show file into paperwork an engineer can actually use: a patch list, a specification sheet and a wiring topology diagram, as PDFs.

It runs in the browser or on the command line. The browser build parses and renders entirely in WebAssembly, so a show file never leaves the machine. There's no upload and no server.

PatchFerret preview

Turn a mixing console show file into a patch list, a specification sheet and a wiring topology diagram.

Your show file never leaves this browser — parsing and PDF generation run locally in WebAssembly.

This is an early preview and very much still in development. Five console families are readable so far, and only partly: channel names and the input patch come through, while head-amp gain, bus sends and processing do not yet. Nothing it produces has been checked against a real console's own screens, and it cannot write or convert show files at all. Treat its output as a draft to check, not as paperwork to hang a show on.

Drop a show file here

or click to choose

› Report header optional · logo, event, venue, engineer…

SHOW

General 1.2.2

Behringer X32 / Midas M32 · v2.7

32	26	80	42	32
Input slots	Reaching a channel	Strips	Outputs	Preamps

DOCUMENTATION

General-1-2-2.pfx.xml	17.3 kB	Download
general-1-2-2-patch-list.pdf	31.0 kB	Download
general-1-2-2-spec.pdf	9.5 kB	Download
general-1-2-2-topology.pdf	7.6 kB	Download

CONVERSION FIDELITY

UNMODELLED `line 16: /config/routing/AES50A`

AES50A output routing blocks [OUT1-8 OUT9-16 OUT1-8 OUT9-16 P161-8 P169-16] are recorded but not expanded into

The hosted tool with the repo's own X32 fixture loaded. It's identified the show as a Behringer X32 / Midas M32 scene, counted its 32 input slots, 80 strips and 42 outputs, and lined up the four documents it makes ready to download, along with the conversion-fidelity list naming the lines of the file it didn't model. Parsed in the browser, and nothing was uploaded.

> **Before you rely on this:** the X32 adapter is written against a real 2,104-line scene file saved by an actual console and is asserted against it in CI. **No output has ever been loaded back into a console**, and **no conversion between consoles exists yet.**
>
> The per-format caveats matter here, and they're in the table below. The Yamaha MMS adapter's **patch-source encoding has never been confirmed against a console's patch screen**, and on TF no connector resolves at all; the tool tells you so rather than guessing. The X32 signal-source enumerations are derived from community documentation and corroborated against real files, not from a running desk.
>
> This codebase was created with AI assistance, directed and reviewed by a human author.

What it reads today

Console	File	Status
Behringer X32 / Midas M32 / X-Air	`.scn`	**Supported** — X-Air not yet tested
Yamaha DM3 / DM7 / TF	`.dm3s` `.tfs` `.dm7s`	**Supported** — names and input patch; head amps and sends not yet
Yamaha CL / QL	`.CLF`	**Supported** — input patch and channel names
Allen & Heath Avantis / dLive	`.tar.gz`	**Supported** — input patch and strip inventory; names and preamps not yet
Allen & Heath SQ	`NVDATA.DAT`	**Supported** — input patch; names and socket *class* not yet identified
DiGiCo, Avid VENUE	—	Planned

```
show file  →  format adapter  →  PFX model  →  ├─ patch list PDF
                                               ├─ specification PDF
                                               ├─ topology PDF
                                               └─ PFX interchange XML
```

Run `info` first, and read the fidelity list

```
patchferret info myshow.scn
```

It prints a summary and, more importantly, **the fidelity list**: everything the adapter read but couldn't carry into the model, which would therefore be lost in a conversion.

That list is the honest measure of how much of your show this tool understands. Read it before you hand a PDF to anybody.

Why the patch list is not obvious

On an X32, getting from an XLR to a fader takes three hops, and all three have to be composed:

1. The routing block maps blocks of **eight physical connectors** onto the 32 **input slots**.
2. The channel config selects which **input slot** feeds each channel, and that's a free mapping.
3. The head-amp table holds the gain for a **connector**, in a flat index across local and both AES50 links.

Assume channel *N* is fed by XLR *N* and you get a patch list that's confident and wrong. In the test fixture, slots 25–32 are AES50-B connectors 1–8 while channels 25–32 carry them, and six channels reach no connector at all.

PatchFerret composes the chain and marks the dead rows rather than inventing a connector for them.

The report header comes from you

Reports carry a header with your logo, the event, the venue, the engineer and so on.

None of that is in a show file (a console stores a mixer state, not the job it was built for), so it comes from a job sheet you supply:

```
patchferret job-template -o job.txt
patchferret report myshow.scn -j job.txt -o ./docs
```

The sheet is `key: value` lines, and **any key it doesn't recognise becomes an extra header field**, so "Truck call" or "Rider rev" work without the tool knowing about them:

```
Event: Summer Live 2026
Venue: Old Granada Studios
Engineer: A. Sargeant
Truck call: 06:00
logo: ./logo.jpg
```

The browser version has the same fields as a form and accepts any image for the logo, converting it on a canvas before handing the bytes over. The CLI takes JPEG, or PNG without transparency. Embedding a transparent PNG would mean decoding pixels, which the dependency-free PDF writer doesn't do on purpose, and it tells you so rather than silently dropping the logo.

In the browser

Module scripts and WebAssembly **won't load from `file://`**, so the local build has to be served over HTTP. The hosted version needs nothing.

If something is wrong

Symptom	Cause
Rows are marked dead	Those channels reach no connector. That is the console's patch, faithfully reported.
A patch list disagrees with the desk	Read the fidelity list from `info` — the adapter may not carry that field yet.
No connector resolves on a TF file	Known: the tool reports it rather than guessing.
The logo was refused	A transparent PNG on the CLI. Use JPEG, or the browser version.
The page will not load locally	It is being opened from `file://`. Serve it over HTTP.